2025 IEEE 49th Annual Computers, Software, and Applications Conference (COMPSAC) | 979-8-3315-7434-5/25/$31.00 ©2025 IEEE | DOI: 10.1109/COMPSAC65507.2025.00192

Quantum Knowledge Graph: Leveraging QNLP for Semantic Relationship Modeling

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***Abstract*—The integration of quantum computing with knowledge graphs presents a transformative approach to intelligent information processing that enables enhanced reasoning, semantic understanding, and large-scale data inference. This study introduces a Quantum Knowledge Graph (QKG) framework that combines Neo4j's LLM Knowledge Graph Builder with Quantum Natural Language Processing (QNLP) to improve the representation, retrieval, and inference of complex knowledge structures. The proposed methodology involves extracting structured relationships from unstructured text, converting them into quantum-compatible representations using Lambeq, and executing quantum circuits via Qiskit to compute quantum embeddings. Using superposition and entanglement, the QKG framework enables parallel relationship processing, contextual entity disambiguation, and more efficient semantic association. These enhancements address the limitations of classical knowledge graphs, such as deterministic representations, scalability constraints, and inefficiencies in the capture of complex relationships. This research highlights the importance of integrating quantum computing with knowledge graphs, offering a scalable, adaptive, and semantically enriched approach to intelligent data processing.**

***Index Terms*—Quantum Knowledge Graphs (QKG), Quantum Natural Language Processing (QNLP), Knowledge Graph(KG), Neo4j, LLM Knowledge Graph Builder**

I. Introduction

Classical knowledge graphs (KGs) have been widely used in various domains such as search engines, recommendation systems, and artificial intelligence (AI) applications based on knowledge graphs. They represent structured relationships between entities using nodes and edges, allowing for efficient data retrieval and reasoning. However, traditional KGs face significant challenges, including their reliance on manually curated data, their deterministic nature, and their inability to effectively model uncertainty. As data sets grow larger, computational bottlenecks arise, making real-time updates and inference increasingly difficult. In addition, classical KGs struggle to encode deeper semantic relationships, leading to limitations in contextual understanding.

To address these limitations, modern graph-based AI systems have integrated large language models (LLMs) with knowledge graph technology. Neo4j's LLM Knowledge Graph Builder is an advanced solution that uses natural language processing (NLP) to automatically extract, enrich, and maintain knowledge graphs. This approach improves scalability by automating entity recognition and relationship mapping while improving contextual accuracy through dynamic updates. However, while LLM-driven KGs improve adaptability and reduce manual efforts, they still operate within classical computing constraints, limiting their ability to process complex, high-dimensional relationships efficiently.

Quantum Knowledge Graphs (QKGs) introduce a novel approach by incorporating quantum natural language processing (QNLP) into knowledge graph construction. Using the principles of superposition and entanglement, QKGs allow for richer semantic modeling and more efficient processing of probabilistic relationships. QNLP enables sentence embeddings that encode contextual meaning with higher fidelity, allowing for improved reasoning and inference. By integrating quantum computing with Neo4j and QNLP frameworks such as Lambeq, QKGs offer a promising future for intelligent systems, enabling enhanced decision making and deeper insights in AI applications.

II. Literature Survey

Neo4j, one of the most prominent graph database platforms, has enabled efficient querying and management of large-scale knowledge graphs. However, despite their utility, traditional knowledge graphs face several limitations.

One major drawback is their deterministic nature, which makes them incapable of effectively representing probabilistic and contextual relationships. Semantic ambiguity and the lack of dynamic adaptability also pose significant challenges. Classical KGs require extensive manual curation and schema-based updates, which make them difficult to scale and adapt in real-time applications. Furthermore, the computational complexity of querying large-scale KGs increases exponentially, leading to inefficiencies in real-time decision-making and reasoning tasks [1].

To address the scalability and adaptability challenges of classical KGs, Neo4j introduced the LLM Knowledge Graph Builder, which integrates Large Language Models (LLMs) with graph-based knowledge representation. This modern approach automates knowledge extraction, relationship enrichment, and dynamic updates, making knowledge graphs more contextually aware and scalable.

2836-3795/25/$31.00 ©2025 IEEE
DOI 10.1109/COMPSAC65507.2025.00192

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Using LLM and NLP techniques, Neo4j enables the creation of contextual and adaptive knowledge graphs. This enhances the ability of AI systems to extract information from unstructured text, resolve ambiguities, and infer implicit relationships between entities. However, despite these improvements, LLM-driven knowledge graphs still operate within classical computing constraints. They struggle with high-dimensional semantic representations and efficient probabilistic reasoning [2][3].

The limitations of both classical KGs and LLM-driven KGs have led to the exploration of Quantum Knowledge Graphs (QKGs). Quantum computing introduces superposition, entanglement, and probabilistic computation, which enable more efficient and scalable knowledge representation [4]. The Quantum Natural Language Processing (QNLP) paradigm applies quantum computing to linguistic structures and knowledge representation, making it a promising solution to overcome the limitations of traditional KGs [5].

Quantum knowledge graphs use quantum state encoding to represent relationships probabilistically, rather than in a deterministic manner. QNLP enables quantum-enhanced embeddings that better capture the contextual nuances of entities and relationships. Using quantum superposition, multiple entity relationships can be processed simultaneously, leading to exponential improvements in query efficiency and knowledge retrieval [6].

Hence, the need for real-time, context-aware, and scalable AI-driven decision-making systems has motivated the shift towards quantum knowledge graphs. Quantum Natural Language Processing (QNLP) provides a powerful framework for modeling relationships probabilistically, allowing AI systems to understand human language [4]. The convergence of quantum computing and AI presents an unique opportunity to redefine intelligent systems, making QKGs a critical advancement in the future of AI, natural language understanding, and knowledge-driven applications.

III. Methodology

This study presents a structured approach to transforming unstructured textual data into a *Quantum Knowledge Graph (QKG)* by utilizing *Neo4j's LLM Knowledge Graph Builder* and *Quantum Natural Language Processing (QNLP)*. The methodology follows a systematic pipeline to ensure efficient extraction, processing, and quantum enhancement of knowledge representations.

A. Data Acquisition and Preprocessing

The first stage involves uploading an unstructured text document (Abhinay Kumar's resume in PDF format) into Neo4j's LLM Knowledge Graph Builder. This document contains various sections such as *education, work experience, projects, and technical skills*. Given that resumes typically follow a semi-structured format, *Neo4j's LLM Knowledge Graph Builder* is utilized to extract entities and relationships from the text, converting it into a structured representation.

B. Knowledge Graph Construction in Neo4j

- Entity Extraction: Using Neo4j LLM Graph Builder, key entities such as names, skills, organizations, projects, and education are extracted.
- Relationship Identification: Relationships between extracted entities are mapped, such as "worked at," "has skill," "contributed to," and "has education".
- Graph Construction: These extracted entities and relationships are stored in Neo4j, forming a classical Knowledge Graph (KG).

C. Transformation of Knowledge Graph Data into Quantum Representations

The next step involves transforming the structured knowledge graph into a format that is compatible with *Quantum Natural Language Processing (QNLP)*. Each entity-relationship pair is represented as a natural language sentence, which serves as input for QNLP.

For example, extracted relationships such as:

- *Abhinay Kumar has education Indian Institute of Technology, Kanpur.*
- *Abhinay Kumar has skill Python.*
- *Abhinay Kumar worked at Gartner.*
- *Abhinay Kumar contributed to Cassava Leave Disease Classification.*

are converted into structured sentences, enabling them to be processed using QNLP techniques.

D. Quantum NLP Processing with Lambeq

Lambeq, a QNLP framework, is then utilized to convert the structured sentences into quantum circuit representations. This conversion involves the use of *DisCoCat (Categorical Compositional Distributional) diagrams*, which model sentence structure in a quantum-native manner.

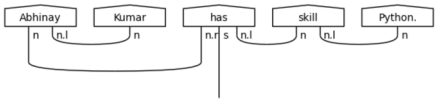


Fig. 1. DisCoCat diagram representation of the sentence "Abhinay Kumar has skill Python."

- Processing Each Sentence: Each structured sentence, derived from the Neo4j knowledge graph, is processed to extract grammatical structures and convert them into a quantum compatible format. The sentence structure is analyzed to identify key elements such as subjects, verbs, objects, and prepositional phrases, which are crucial for knowledge representation in a quantum system.
- Parsing the Sentence into a DisCoCat Diagram: Using Lambeq's BobcatParser, the sentence is parsed into a DisCoCat diagram, which represents the syntactic structure of the sentence using category theory and tensor networks. The DisCoCat diagram provides a foundation for quantum circuit generation by mapping words to

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tensor representations, allowing for a quantum-native way to encode linguistic meaning.

- DisCoCat diagram : Fig. 1. represents the quantum linguistic structure of the sentence ”Abhinay Kumar has skill Python,” mapping its grammatical dependencies to tensor-based quantum representations. Captures context-sensitive information, making it useful for knowledge graph embeddings. Each connection represents a quantum interaction that helps in semantic reasoning. Each word is assigned a quantum type, where n (noun) represents entities, s (sentence type) denotes a complete statement, and n.l/n.r indicate dependencies linking words. ”Abhinay” (n) and ”Kumar” (n) are both nouns. ”Kumar” has an additional left-adjoint (n.l), linking it to ”Abhinay”. This noun phrase (”Abhinay Kumar”) is treated as a single entity in the knowledge graph. ”has” is a verb connecting ”Abhinay Kumar” to ”skill”. ”has” takes two arguments: a subject (”Abhinay Kumar”, n.r) and an object (”skill”, n.l) The output of ”has” is a sentence (s), meaning it completes a full meaningful relationship. ”skill” (n) is modified by ”Python” (n). ”Python” has an adjective-like dependency (n.l), linking it to ”skill”. This phrase establishes a relationship between ”skill” and ”Python”, treating ”Python” as an attribute of ”skill”.
- Converting the DisCoCat Diagram into a Quantum Circuit: Once the sentence is converted into a DisCoCat diagram, it is then mapped to a quantum circuit. This conversion follows an ansatz-based approach, where each atomic type (such as nouns, verbs, and sentences) is assigned a specific quantum representation. Each word in the sentence contributes to the quantum state, and the circuit is generated based on grammatical dependencies.
- Quantum Circuit Construction : A quantum circuit is then constructed on the basis of the converted DisCoCat representation. The quantum circuit follows a superposition-based encoding scheme, where different grammatical elements interact through quantum gates.

E. Quantum Circuit Execution and Embedding Storage

Once the quantum circuits are generated, they are executed on a quantum simulator backend using *Qiskit*. The results of the quantum measurement are then normalized and stored in *Neo4j*, where each sentence is stored as a node labeled Quantum Sentence in Neo4j. Quantum Embeddings are stored as properties of these nodes.

The quantum embeddings improve knowledge representation by introducing:

- Context-aware entity relationships
- Probabilistic reasoning for ambiguous connections
- Enhanced semantic similarity detection

F. Visualization of the Quantum Knowledge Graph

The final step involves visualizing the *Quantum Knowledge Graph (QKG)* to better understand the relationships formed between entities. *NetworkX* and *Matplotlib* are used to create a graphical representation of the QKG. NetworkX is used to visualize the Quantum KG. Edges are formed based on cosine similarity between quantum embeddings. Nodes represented quantum-processed sentences, showcasing semantic relationships.

This methodology effectively integrates quantum computing principles with knowledge graph technology leveraging quantum-enhanced representations for improved decision-making and semantic understanding.

IV. Preliminary Results and Analysis

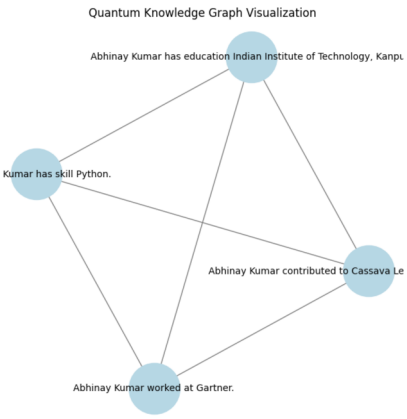


Fig. 2. Quantum Knowledge Graph Visualization

Fig 2 illustrates the structured relationships extracted from unstructured text and enhanced using quantum embeddings. The QKG effectively captures complex semantic relationships by encoding probabilistic quantum states into the knowledge graph, providing richer context-aware reasoning than classical methods.

A. Quantum Knowledge Graph Representation

The QKG visualization presents a network of entities and relationships derived from Abhinay Kumar's resume. Key relationships identified include Abhinay Kumar has education at Indian Institute of Technology, Kanpur , Abhinay Kumar worked at Gartner, Kumar has skill Python, Abhinay Kumar contributed to Cassava Leaf Disease Classification. Each entity and relationship has been enhanced using quantum embeddings, enabling improved semantic inference and contextual association compared to traditional knowledge graphs. The QKG captures semantic relationships between education, skills, work, and projects.

Unlike classical knowledge graphs that rely on fixed relationships, QKGs encode entity relationships as quantum states, allowing probabilistic connections between skills, education, and work experience. Example: If multiple candidates with similar education and work history also have expertise in AI, QKG may infer that Abhinay Kumar might have AI-related expertise even if it is not explicitly mentioned. QKG can predict missing relationships by analyzing semantic similarities between quantum embeddings. Example: If most Gartner employees with a background in Python also possess expertise

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in Machine Learning, QKG might suggest a probable link between Abhinay Kumar and Machine Learning, even if it is not explicitly listed.

B. Quantum Knowledge Graph Analysis

Quantum Knowledge Graph approach demonstrates several key advantages over classical knowledge graph methods:

Traditional knowledge graphs store relationships in a deterministic manner, making it difficult to infer implicit connections between entities. The QKG leverages Quantum Natural Language Processing (QNLP)-based embeddings, where sentences are encoded into quantum states, preserving semantic richness and contextual relationships. Quantum embeddings are computed through quantum circuits that capture higher-order dependencies, allowing the model to connect seemingly unrelated entities through probabilistic inference.

Traditional knowledge graphs traverse edges sequentially, making query execution expensive as the graph size grows. Quantum knowledge graphs leverage quantum superposition, allowing multiple relationships to be evaluated in parallel within a single quantum operation. The use of quantum circuits to encode relationships enables parallel evaluation of multiple entity associations, significantly reducing retrieval latency. Instead of querying individual relationships one by one, quantum states allow simultaneous evaluation of multiple paths, accelerating response time.

C. Quantum Knowledge Graph Scalability

QKGs inherently support quantum parallelism which provides a distinct advantage in scalability: as the graph size increases, QKGs maintain inference efficiency, unlike classical KGs, which face latency due to sequential edge traversal.

V. Future Work and Directions

The implementation of Quantum Knowledge Graphs (QKGs) has demonstrated significant improvements in knowledge representation and reasoning. However, several key areas require further exploration and optimization to fully harness the potential of quantum computing for intelligent systems. Currently, quantum simulations are executed on quantum simulators (Qiskit Aer backend) due to hardware limitations. The next step is to transition from simulated quantum circuits to execution on real quantum processors (e.g., IBM Quantum). Optimizing QNLP models for near-term quantum devices (NISQ systems) to mitigate the effects of quantum noise and decoherence. Developing noise-resilient quantum embeddings using techniques like error correction codes and quantum kernel methods to improve robustness in real-world scenarios.

VI. Conclusion

The Quantum Knowledge Graph (QKG) framework outperforms classical and LLM-driven knowledge graphs by leveraging quantum superposition and entanglement for parallel relationship processing, significantly reducing query execution time from $O(N^2)$ in classical methods to $O(\log N)$. Unlike deterministic classical KGs and limited LLM-KGs, QKGs utilize quantum embeddings to capture probabilistic relationships, enabling enhanced semantic understanding and improved entity disambiguation. The scalability allows QKGs to process high-dimensional knowledge structures efficiently, making them ideal for AI-driven applications such as intelligent search engines and automated decision-making.

Quantum Knowledge Graphs further enhance information retrieval by leveraging semantic similarity from quantum embeddings. Unlike classical searches that rely on explicitly defined relationships, QKGs infer hidden connections through quantum embeddings, enabling AI-driven systems to be more intelligent, scalable, and contextually aware. In education platforms, QKGs facilitate personalized learning by recommending courses based on users' skills, while in recruitment, they identify candidates with transferable competencies, improving talent matching and career guidance. AI assistants benefit from context-aware knowledge retrieval, leading to more accurate, dynamic, and intelligent responses in search engines and chatbots. As quantum computing advances, QKGs play a crucial role in intelligent systems, driving improvements in semantic reasoning, knowledge discovery, and AI-driven decision-making.

Acknowledgement

The work is partially supported by the U.S. National Science Foundation under awards #2433800, #2413540 and #2100115.

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